# Internship Report

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KM Knitwear Private Ltd.,
City Garden, 14-D, E, F Lakshmi Nagar,
1st St, Tiruppur - 641602

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# 1. Introduction

**Summary ;**

This internship report provides an in-depth analysis of the operations at K.M Knitwear, a prominent garment manufacturing company based in Tirupur, Tamil Nadu. The primary objective of this internship was to gain practical exposure to the intricacies of the garment manufacturing process, from order inception to product delivery.

**Purpose of The Internship ;**

The primary purpose of this internship at K.M Knitwear is to provide a comprehensive understanding of the garment manufacturing process, from order conception to final product delivery. By immersing in the day-to-day operations of the company, as an intern, I gained practical experience in various aspects of the textile industry, including but not limited to:

a. Developing professional network
b. Contributing to operational efficiency
c. Building industry knowledge
d. Enhancing problem-solving abilities
e. Developing analytical skills
f. Understanding the production process

**Acknowledgement ;**

I express sincere gratitude to The people of K.M Knitwear for their invaluable guidance and support. Their mentorship was instrumental in my learning. I am thankful to the entire K.M. Knitwear team for their cooperation and assistance.

# 2. Company's Profile

### Company Overview ;

**Name**: KM Group of Companies

**Founded**: Over 25 years ago

**HeadOffice**: 14-D, E, F Lakshmi Nagar,
1st St, Tiruppur - 641602

**Industry**: Textile Manufacturing

### Core Business ;

The KM Group specializes in textile manufacturing, with a focus on garment manufacturing. The company operates as a fully vertical conglomerate, managing all aspects of production from raw materials to finished products.

### Key Strengths ;

i. Fully vertical integration
ii. Environmental sustainability
iii. Customized, high-quality products
iv. Long-term client relationships
v. Future-ready infrastructure

### Client Focus ;

The KM Group caters to quality-conscious clients seeking a committed vendor partner. The company specializes in developing niche, high-quality products tailored to customer specifications.

### Manufacturing Capabilities ;

- End-to-end production from raw materials to finished goods
- Customization according to client designs

# 3. My Learnings

As a mechanical engineer, transitioning into the textile industry might seem unconventional. However, my internship at K.M Knitwear proved to be a valuable experience that bridged the gap between these two seemingly disparate fields. By gaining a unique perspective and developing a comprehensive understanding of the production process. This section will delve into the specific ways in which my engineering background contributed to my learning and growth during my internship.

**Understood the Manufacturing Process:** Gained a comprehensive understanding of the entire infant garment production cycle, from raw material to finished product.

**Developed Technical Proficiency:** Acquired experience with machinery and equipment used in their operation.

**Enhanced Problem-Solving Skills:** Developed the ability to identify and resolve production challenges.

**Improved Quality Focus:** Strengthened understanding of quality control standards and implemented measures to enhance product quality.

**Time Management and Organization:** By managing time effectively for various tasks and deadlines in a fast-paced setting.

**Expanded Industry Knowledge:** Gained insights into market trends, customer requirements, and industry best practices.

# 4. Operations Involved

## I. Pre-Production ;

As a garment export house, The pre-production process mainly involves,

**1. Meeting with Buyers;** The apparel manufacturing process commences with order acquisition from buyers. This involves presenting design concepts, receiving buyer specifications, and negotiating order details. Close collaboration between designers, merchandisers, and buyers is crucial in this stage to ensure alignment with market trends and customer preferences.

**2. Development of Initial Samples for the Buyer;** Upon receiving the buyer's concept or instructions for a new style, samples are created using available fabrics and trims. The sampling team develops these samples under the supervision of the sampling merchandiser. If necessary, discussions regarding new requirements for raw fabrics, trims, and garment samples are conducted with the buyer. Buying houses often closely monitors the new sample development process.

**3. Development of Fabric Samples, Bit Loom, Print, and Embroidery Artwork;** Fabrics are developed according to the buyer's specifications. This includes sourcing customer-specific fabrics with matching properties, dyeing, and finishing for solid colors. The approval process for solid color lap dips is crucial and continues until the sample is approved. For yarn-dyed fabrics, merchants develop fabric samples with specified designs, such as stripes or checks, known as Bit Looms. Other approvals, such as print and embroidery artwork and color approvals, may occur at a later stage during pre-production.

**4. Costing of a Garment (Complete Cost as Well as Manufacturing Cost);** Merchants prepare cost sheets detailing the cost breakdown, including raw material costs, manufacturing costs, overheads, and margins. Costing is a critical stage, as it determines whether the company secures the order. If the garment cost is too high, the manufacturer may lose the order; if the cost is too low, the factory may not earn a profit. Accurate cost estimation should be data-driven.

**5. Pattern Making, Correction of Pattern, and Pattern Grading;** A pattern master in the factory prepares the initial fit pattern, then revises it based on buyer comments and adjustments to the fit sample. After fit approval, the pattern master grades the pattern for size set samples in specified sizes. When the order is ready for production, patterns are graded for the entire size range. Garment factories typically use CAD systems in addition to manual pattern drafting and grading.

**6. Fit Sample, Size Set Sample Making, and Approval from Buyer;** Each sample serves a specific purpose. Samples are created in the sampling department and sent to the buyer for approval. The designer and the relevant merchandiser from the buyer's side review the garment samples against the design and tech pack, providing feedback on the submitted samples.

**7. Correction of Fit Samples According to Buyer Comments;** If the sample is not approved or requires further work as recommended by the buyer, corrections are made and the sample is resubmitted for approval.

**8. Sample Approval Process;** During this stage, sample approval is obtained from the buyer. This includes approval of fabric swatches, print colors, embroidery designs, beadworks, and other processes. Sometimes, factories also need to get trim cards approved.

**9. Production Planning, Material Planning, and Line Planning;** Effective planning is essential to start production on time and ensure timely shipment of orders. This involves planning for material sourcing, production capacity, and line planning. Job scheduling and responsibilities are defined at this stage.

**10. Development of Fabrics, Trims, Accessories, and Packing Materials;** Sourcing of raw materials, including fabrics, trims, and accessories, is conducted during this stage.

**11. Testing of Fabrics and Other Raw Materials;** The Physical properties of bulk fabrics are tested in in-house testing labs to ensure quality.

**12. Study of Approved Sample;** For operation breakdown, calculating work content, and identifying critical operations and line setting, the technical team and industrial engineers study the approved garment sample meticulously.

## II. Bulk-Production ;

### 1. Spinning;

The spinning division is known for its innovative infrastructure and seamless production process. The division excels in manufacturing error-free and clean yarn, with a production capacity exceeding 60 tons per day. It prides itself on unmatched production quality, endorsed by relevant certifications, and maintains consistent quality measures throughout the entire process, from cotton mixing to yarn production.

The process begins with selecting the finest cotton mixes to ensure high-quality yarn. Advanced machinery, including clearers, ensures 100% defect-free cotton, while comber machines produce superior quality yarn. High-speed ring frames and autoconers facilitate seamless yarn manufacturing, with automation ensuring consistency and precision.

## 2. Knitting;

Following the spinning process, the required quantity of yarn is calculated based on fabric specifications like GSM and finish. Subsequently, the yarn is transformed into fabric within the knitting division. The advanced in-house machinery, a testament to innovation and excellence, allows for the creation of diverse finishes including Rib, Jersey, Airtex, Honeycomb, Lycra Jersey, Fleece, and Mini Jacquards, tailored to any specified form or structure.

## 3. Dyeing;

A dyeing facility That incorporates advanced soft-flow and air-flow technology, enabling precise temperature control during the dyeing process. The computerized laboratory, equipped with data color-matching capabilities, ensures consistent color accuracy. Where The laboratory is staffed by experienced chemists.

To achieve optimal fabric finish, the company utilizes a compacting unit. This process effectively controls fabric shrinkage, spirality, and grammage. The compacting machinery, sourced from the USA and Italy, imparts a superior finish to the fabric.

## 4. Cutting;

The cutting process in a garment manufacturing factory is a critical stage that transforms fabric rolls into garment pieces ready for assembly. This process ensures precision and efficiency, laying the foundation for the final garment.

**a. Fabric Inspection and Preparation;** Before cutting begins, the fabric undergoes a thorough inspection to check for defects, color consistency, and overall quality. The fabric is then prepped, which includes spreading (layering) it onto cutting tables. For large-scale operations, fabric is spread using automated machines that layer the fabric to the required number of plies, based on the garment design and size specifications.

**b. Pattern Layout;** Patterns are placed on the spread fabric according to the garment design. Patterns are often digitized and input into computer-aided design (CAD) systems, which help in optimizing the layout to reduce fabric waste. For manual cutting, pattern pieces are placed on the fabric in a way that maximizes the use of material and minimizes waste.

**c. Cutting;** The actual cutting process can be performed using various methods depending on the factory's setup:

- **Manual Cutting**: Skilled cutters use scissors or rotary cutters to cut the fabric according to the laid-out patterns. This method is less common in high-volume production due to its time-consuming nature.
- **Automatic Cutting**: For higher efficiency, automated cutting machines such as fabric cutters or CNC (Computer Numerical Control) machines are used. These machines can cut multiple layers of fabric simultaneously with high precision.

**d. Fabric Segregation;** Once cut, the fabric pieces are carefully separated and labeled. Each piece is sorted based on the garment size and style to ensure correct assembly in subsequent stages. Proper segregation and labeling prevent mix-ups and ensure that each garment piece is accurately matched.

**e. Quality Check;** After cutting, a final quality check is performed to ensure that all pieces are accurately cut and free from defects. This step helps to identify any issues that might have arisen during the cutting process, such as irregularities or inaccuracies.

### 5. Printing;

**a. Design Preparation;** The process begins with the creation of the design that will be printed on the fabric. This design is often developed using graphic design software and is finalized based on client specifications or seasonal trends.

**b. Screen or Plate Making;** For screen printing, screens are prepared with the design pattern. Each color in the design requires a separate screen. In the case of digital or rotary screen printing, plates or digital files are created instead. These screens or plates serve as templates for applying the ink to the fabric.

**c. Ink Preparation;** The appropriate inks are selected based on the fabric type and the desired finish. The inks are then mixed to achieve the exact colors required for the design. Special inks, such as those for metallic or reflective effects, are also prepared if needed.

**d. Printing;**

- **Screen Printing**: In this traditional method, ink is pushed through the prepared screens onto the fabric. Each color is applied in separate layers using different screens, which are aligned precisely to ensure accurate color registration.
- **Digital Printing**: Digital printers apply ink directly onto the fabric from digital files, allowing for detailed designs and a wide range of colors. This method is efficient for small runs or intricate patterns.
- **Rotary Screen Printing**: This continuous process uses cylindrical screens and is ideal for high-speed production. The fabric is fed through rotating screens where the ink is applied, ensuring consistent and high-quality prints.

**e. Drying and Curing;** After printing, the fabric must be dried or cured to set the ink. This is typically done using heat. For screen printing, a heat press or conveyor dryer is used to ensure that the ink adheres properly and becomes durable. Digital and rotary screen prints also undergo a curing process to fix the ink.

**f. Quality Control;** The printed fabric undergoes quality checks to ensure that the colors are accurate and the print quality meets the required standards. This includes inspecting for any inconsistencies, defects, or color mismatches.

**g. Finishing;** Finally, the fabric is finished to achieve the desired look and feel. This may include additional treatments such as washing, softening, or applying protective coatings to enhance the print's durability and appearance.

### 6. Embroidery;

The embroidery process starts with designing patterns or logos, which are then turned into digital files. These designs are uploaded to an embroidery machine.

First, the fabric is secured in a hoop to keep it in place. The machine follows the digital design and stitches the pattern onto the fabric using different colored threads. The machine is closely monitored to ensure the design is stitched correctly.

Once the embroidery is complete, the garment is checked for any issues with the stitching. If needed, adjustments are made. Finally, the garment is steamed and pressed to remove any marks and give it a clean, finished look before it is packaged and prepared for shipping.

### 7. Sewing;

The sewing process is where the fabric pieces come together to form the finished garment. The process starts with setting up sewing lines and assigning tasks to workers. The sewing section is divided into different workstations, each responsible for specific parts of the sewing process.

The first step involves laying out the cut fabric pieces according to the garment's design. These pieces are then fed into sewing machines, where operators carefully stitch them together based on detailed patterns and instructions. Different machines are used for various tasks; for example, overlock machines finish the edges, while straight stitch machines handle the main seams.

Once the pieces are stitched, additional tasks are carried out, such as attaching zippers, buttons, and labels if available. Quality control is an ongoing part of the process. Inspectors check the garments at various stages to ensure they meet the required standards and fix any issues.

### 8. Finishing;

The final phase of the production process is ensuring that each garment meets the buyer's vision through meticulous checking, ironing, and packaging. The experienced quality control team conducts an in-depth final product inspection, adhering to KM-certified standards and globally accepted AQL (Acceptable Quality Level) standards. This rigorous inspection is performed despite comprehensive end-to-end quality checks at every stage of the manufacturing process, ensuring that each garment is error-free and contamination-free.

In this phase, garments undergo thorough inspections to identify and rectify any defects or discrepancies. They are then subjected to ironing to achieve a polished, wrinkle-free appearance, which is essential for presentation and quality assurance. Following ironing, garments are run through metal detection machines to ensure the complete removal of any metallic contaminants, safeguarding the product's integrity. Finally, garments are packaged in metal-free zones.

### 9. Packing;

Once the garments are completed and inspected, they are neatly folded and packed in custom-made corrugated boxes. Each box is labeled with relevant shipping information and barcodes to facilitate smooth handling and tracking. The boxes are then securely sealed to prevent any damage or contamination, and ready for shipment.

# 5. Safety

The company prioritizes the safety and well-being of its employees. To mitigate potential risks, several safety measures have been implemented throughout the facility.

**Fire Safety:** A comprehensive fire safety system is in place. Fire extinguishers of appropriate types are strategically located across the factory. Regular inspections and maintenance of these extinguishers are conducted to ensure their operational readiness. Emergency exits are clearly marked and regularly checked for obstructions. Fire drills are conducted periodically to familiarize employees with emergency procedures.

**First Aid:** Well-stocked first aid boxes are positioned at accessible locations within the factory. Employees are trained in basic first aid to provide immediate assistance in case of minor injuries. Additionally, the company maintains a dedicated medical room equipped with essential medical supplies for treating more serious injuries.